

SYSTEM ADMINISTRATION & MAINTENANCE

Reviewer + Exam-Like Questionnaire

Based on Week 1: Introduction to System Administration and Network Administration and Lesson 2: Operating System Installation and Configuration.

PART I — INTRODUCTION TO SYSTEM & NETWORK ADMINISTRATION

System Administration: Management, maintenance, and support of computer systems, servers, and IT infrastructure to ensure reliable and secure operation.

Key responsibilities: Install operating systems; create user accounts; manage permissions; perform backups; apply updates and patches; troubleshoot system problems.

Memory: System Admin = Systems / Servers / Software

Network Administration

Focuses on managing network devices and communication between systems.

Key responsibilities: Configure routers and switches; manage IP addressing; monitor network performance; maintain connectivity; implement network security; troubleshoot network issues.

Memory: Network Admin = Network Devices / IP / Connectivity

System Administrator vs Network Administrator

System Administrator	Network Administrator
Manages servers	Manages routers & switches
Creates users	Configures IPs
Maintains operating systems	Configures VLANs
Performs backups	Monitors traffic
Installs applications	Configures routing

Enterprise Network Overview

Internet → Router → Switch → PCs / Servers

Router: connects different networks.

Switch: connects devices in a LAN.

Firewall: filters and protects traffic.

Server: provides services to clients.

Why Learn Cisco?

Cisco technologies are widely used in enterprises. Benefits include industry-recognized skills, hands-on networking experience, preparation for CCST / CCNA, better employability, and understanding real-world networks.

Career opportunities: System Administrator, Network Administrator, IT Support Engineer, Cloud Administrator.

PART II — OPERATING SYSTEM INSTALLATION & CONFIGURATION

Hardware Assessment

Before installation, confirm CPU speed, RAM capacity, and disk space meet OS requirements. Cross-reference hardware with manufacturer specifications to avoid driver and firmware conflicts. Prepare bootable USB/DVD media or configure a network PXE server.

Pre-Installation: Three Critical Decisions

- 1. Boot Sequence:** Configure BIOS/UEFI to prioritize installation media.
- 2. Installation Strategy:** Choose Clean Install, Upgrade, or Multi-boot.
- 3. Deployment Environment:** Decide between physical hardware and virtualization.

Disk Partitioning

Partitioning divides a physical disk into logically independent volumes. **MBR** is legacy, has a 2 TB limit, and supports up to four primary partitions. **GPT** is modern and supports larger disks. An extended partition can contain multiple logical partitions. A swap area is a dedicated partition used for virtual memory.

File Systems & Formatting

Formatting: prepares a partition to store system files.

NTFS: default for Windows; supports encryption, compression, and large volumes.

ext4: standard for Linux; reliable, journaling-enabled, and widely adopted.

Installation Process

- 1. Launch Installer:** Windows setup.exe or Anaconda on Linux distributions.
- 2. Regional Settings:** language, time zone, keyboard layout, and regional preferences.
- 3. Licensing:** enter product key and verify software licensing.

Initial System Configuration

Network: configure IP addressing, DNS, and firewall/security policies.

System Services: enable essential services and set startup parameters.

Startup Optimization: use msconfig (Windows) or systemd (Linux) to disable unnecessary background tasks and improve boot time.

User & Group Account Management

Principle of Least Privilege: users should only have the access they need.

Administrator: full system control; use sparingly and protect with strong credentials.

Standard User: limited privileges reduce risk.

Guest: temporary, highly restricted access; should be disabled in production.

User Groups: organize users to manage permissions efficiently.

Maintenance & System Security

Updates & Patches: install critical OS updates to close known vulnerabilities.

System Logs: monitor kernel events, service failures, and security incidents.

Admin Tools: Device Manager and Disk Cleanup help maintain hardware health and free storage.

System Lifecycle

- 1. New:** freshly installed, fully patched, optimally configured.
- 2. Configured:** customized for production; routine defragmentation and error checking applied.
- 3. Degraded:** performance declining; verify backups and assess hardware for replacement.
- 4. Renewed:** re-imaged, upgraded, or replaced to restore integrity and performance.

Memory: New → Configured → Degraded → Renewed

HIGH-YIELD EXAM POINTS

- System Admin → servers, OS, users, backups.
- Network Admin → routers, switches, IPs, VLANs, routing.
- Router → connects different networks.
- Switch → connects devices in a LAN.
- Firewall → filters/protects traffic.
- Server → provides services to clients.
- MBR → legacy, 2 TB limit, up to 4 primary partitions.
- GPT → modern, supports larger disks.
- NTFS → Windows.
- ext4 → Linux.
- Least Privilege → only necessary access.
- Administrator → full control.
- Standard User → limited privileges.
- Guest → temporary/restricted; disable in production.
- Windows startup tool → msconfig.
- Linux system/service manager → systemd.
- Lifecycle → New → Configured → Degraded → Renewed.

EXAM-LIKE QUESTIONNAIRE

Part A — Multiple Choice

1. What is the primary purpose of system administration?

- A. Designing websites
- B. Managing, maintaining, and supporting computer systems and IT infrastructure
- C. Creating mobile applications
- D. Designing network cables

2. Which task is primarily performed by a system administrator?

- A. Configuring VLANs
- B. Configuring routing
- C. Creating user accounts
- D. Monitoring network traffic

3. Which task is primarily associated with a network administrator?

- A. Installing applications
- B. Performing backups
- C. Creating user accounts
- D. Configuring routers and switches

4. Which device connects different networks?

- A. Switch
- B. Router
- C. Server
- D. Firewall

5. Which device connects devices within a LAN?

- A. Router
- B. Server
- C. Switch
- D. Firewall

6. Which device filters and protects network traffic?

- A. Switch
- B. Server
- C. Router
- D. Firewall

7. Which device provides services to clients?

- A. Server
- B. Switch
- C. Router
- D. Firewall

8. Which sequence represents the basic enterprise network overview?

- A. Switch → Internet → Router → PCs
- B. Internet → Router → Switch → PCs/Servers
- C. Internet → Switch → Router → PCs
- D. Router → Internet → Switch → Servers

9. Before installing an operating system, which should be checked?

- A. CPU, RAM, and storage
- B. Keyboard color
- C. Monitor size only
- D. Wallpaper resolution

10. What should be configured to prioritize installation media during boot?

- A. Firewall
- B. BIOS/UEFI
- C. Device Manager
- D. Disk Cleanup

11. Which is NOT an installation strategy mentioned?

- A. Clean Install
- B. Upgrade
- C. Multi-boot

D. Dual Firewall

12. Which partitioning scheme is modern?

- A. MBR
- B. GPT
- C. FAT
- D. NTFS

13. What is the MBR disk size limit stated in the lesson?

- A. 500 GB
- B. 1 TB
- C. 2 TB
- D. 4 TB

14. How many primary partitions can MBR support according to the lesson?

- A. 2
- B. 4
- C. 8
- D. Unlimited

15. What is the purpose of an extended partition?

- A. It provides internet access
- B. It contains multiple logical partitions
- C. It encrypts the entire disk
- D. It replaces the BIOS

16. Which file system is the default/common choice for Windows?

- A. ext4
- B. NTFS
- C. GPT
- D. MBR

17. Which file system is standard for Linux?

- A. NTFS
- B. FAT32
- C. ext4
- D. MBR

18. What does formatting do?

- A. Connects a computer to the internet
- B. Prepares a partition to store system files
- C. Creates a user account
- D. Configures a router

19. Which is an example of installation media?

- A. Bootable USB

- B. Mouse
- C. Keyboard
- D. Monitor

20. Which Windows tool can be used to optimize startup?

- A. systemd
- B. Anaconda
- C. msconfig
- D. BIOS

21. Which Linux component is mentioned for managing system startup/services?

- A. msconfig
- B. systemd
- C. Device Manager
- D. setup.exe

22. What does the Principle of Least Privilege mean?

- A. Everyone should have administrator access
- B. Users should only have the access they need
- C. Users should never have passwords
- D. Guests should have full access

23. Which account has full system control?

- A. Guest
- B. Standard User
- C. Administrator
- D. Group User

24. Which account has limited privileges to reduce risk?

- A. Administrator
- B. Standard User
- C. Root Router
- D. Guest Administrator

25. What should happen to a Guest Account in production?

- A. Give it administrator access
- B. Make it the default account
- C. Disable it
- D. Use it for backups

26. Why are user groups useful?

- A. They increase CPU speed
- B. They organize users for efficient permission management
- C. They replace operating systems
- D. They configure routers

27. Why should critical OS updates be installed?

- A. To change the wallpaper
- B. To close known vulnerabilities
- C. To increase monitor resolution
- D. To create VLANs

28. What can system logs help monitor?

- A. Kernel events and service failures
- B. Monitor brightness
- C. Keyboard layout only
- D. CPU brand

29. Which tool is mentioned for maintaining hardware health?

- A. Device Manager
- B. Router Manager
- C. CCNA
- D. Anaconda

30. What is the correct system lifecycle order?

- A. New → Degraded → Configured → Renewed
- B. Configured → New → Renewed → Degraded
- C. New → Configured → Degraded → Renewed
- D. Renewed → New → Configured → Degraded

Part B — Identification

- 31. The management and maintenance of computer systems, servers, and IT infrastructure.
- 32. The administration role focused on network devices and communication between systems.
- 33. A device that connects different networks.
- 34. A device that connects devices within a LAN.
- 35. A device that filters and protects traffic.
- 36. The modern disk partitioning scheme that supports larger disks.
- 37. The legacy partitioning scheme with a stated 2 TB limit.
- 38. The Windows file system discussed in the lesson.
- 39. The Linux file system discussed in the lesson.
- 40. The principle where users receive only the access they need.
- 41. The Windows tool mentioned for optimizing startup.
- 42. The Linux system component mentioned for startup/service management.
- 43. The lifecycle phase where system performance is declining.
- 44. The lifecycle phase where a system is re-imaged, upgraded, or replaced.
- 45. A dedicated partition used for virtual memory.

Part C — True or False

- 46. A system administrator primarily configures routers and switches.
- 47. A network administrator may manage IP addressing.
- 48. A switch connects devices in a LAN.
- 49. A router connects different networks.
- 50. A firewall filters and protects traffic.
- 51. GPT is described as a legacy partitioning scheme.
- 52. MBR has a 2 TB limit according to the lesson.
- 53. NTFS is associated with Windows.
- 54. ext4 is a standard Linux file system.
- 55. The Principle of Least Privilege gives every user full administrator access.
- 56. Guest accounts should be disabled in production environments.
- 57. System logs can help identify service failures.
- 58. Critical updates can help close known vulnerabilities.
- 59. A degraded system has declining performance.
- 60. The final lifecycle phase listed is Renewed.

Part D — Scenario-Based Questions

- 61. Employees cannot log into their Windows accounts. Which administrator should investigate first, and why?

62. The office cannot access the internet because the router is misconfigured. Which administrator should handle the problem?
63. A company needs to install Windows Server on a new machine. Which role is primarily responsible?
64. Users cannot communicate between VLAN 10 and VLAN 20. Which administrator should investigate?
65. A server's hard drive must be backed up before maintenance. Which administrator's responsibility is this?
66. A switch suddenly stops forwarding network traffic. Which administrator should investigate?
67. You are installing an OS from a USB, but the computer boots directly into the existing OS. What should you check?
68. A user asks for administrator privileges even though they only need to use a basic application. What security principle should guide your decision?
69. A computer has declining performance, and its backups have been verified. Which system lifecycle phase is it in, and what should be considered next?
70. A university suddenly loses internet access. List the basic troubleshooting steps/devices you would check based on the concepts in the lessons.

ANSWER KEY

Part A — Multiple Choice

1-B 2-C 3-D 4-B 5-C 6-D 7-A 8-B 9-A 10-B
11-D 12-B 13-C 14-B 15-B 16-B 17-C 18-B 19-A 20-C
21-B 22-B 23-C 24-B 25-C 26-B 27-B 28-A 29-A 30-C

Part B — Identification

31. System Administration
32. Network Administration
33. Router
34. Switch
35. Firewall
36. GPT
37. MBR
38. NTFS
39. ext4
40. Principle of Least Privilege
41. msconfig
42. systemd
43. Degraded
44. Renewed
45. Swap Area

Part C — True or False

46. False
47. True
48. True
49. True
50. True
51. False
52. True
53. True
54. True
55. False
56. True
57. True
58. True

59. True

60. True

Part D — Scenario Answers

61. System Administrator — the issue involves user accounts/OS access.

62. Network Administrator — the problem involves a router and network connectivity.

63. System Administrator — installing and configuring an operating system is a system administration task.

64. Network Administrator — VLANs and network communication fall under network administration.

65. System Administrator — backups are a listed system administration responsibility.

66. Network Administrator — a switch is a network device.

67. Check the BIOS/UEFI boot sequence and ensure the USB installation media has priority.

68. Principle of Least Privilege — give the user only the permissions required for their work.

69. Degraded — performance is declining; assess hardware for possible replacement after verifying backups.

70. Check connectivity; check the router and switch; determine whether one or many computers are affected; check IP/network configuration; identify where communication fails; if the router is faulty, consider an appropriate temporary solution while awaiting repair/replacement.